CS11013 report 31

Group Members:

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- Abdul Rashid (25384001)
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Abstract:

This project aims to read in data from CSV files containing flight information, manipulate them and display them visually through charts and texts using Processing. We planned to have a filtering system and user interactivity features. We achieved these with a search and scroll system, and by providing users with tools such as dropdown boxes and sliders. The system efficiently handles small to medium sized datasets, while keeping the potential for larger ones with all the necessary information processed, keeping the interface clean and straightforward.

Literature review:

https://forum.processing.org/one/topic/movement-using-arrow-keys.html
https://processing.org/examples/scrollbar.html
-This project used some guidance from the Processing Forum for implementing arrow key movement, and used the scrollbar example for developing the slider functionality

Project details:

#Stakeholders-

We built this project with the aim of helping airlines and data analysts gain insights from flight data to improve efficiency, such as improving flight scheduling, reducing delays, managing cancellations and routes, etc.

#Problem-

Airlines and analysts work with large amounts of flight data in .csv format, which is difficult to understand and analyze quickly. It's important that they're able to identify patterns in flight delays, cancellations, and others, but it's not easy to identify these patterns from just the raw csv file.

#Solution-

As a solution, we developed a dashboard that converts the raw, hard-to-read flight data into visual charts. It includes filtering options to allow users to analyze the data dynamically.

#*Features-*

- CSV data processing
- Three screens for different data views: Pie charts, Search, and Bar chart screens
- Navigation between screens
- Visual representation of data: pie charts and bar charts
- User inputs to filter the data and change the chart dynamically
    - Text field
    - Slider
    - Multi-handle slider
    - Dropdown
    - Multi-select dropdown
- Additional metrics

#*How it works-*

The program reads the data from a CSV file, in our case “flights2k.csv”, processes it, creates an instance of a Flight class for each entry, and stores each instance in an ArrayList. We have a helper class, Flights, which takes in the ArrayList of flights created when the program starts. The purpose of this class is to help with filtering the data and to create the results for charts.

The screen then diverges into three different screens, each showing different charts and data filtering options. These are controlled by three buttons. To make sure that only one screen is showing up at a time, each screen is assigned a code (0,1,2), and only one code would be active depending on which button is pressed. Some skeleton functions were also added to certain screens to make sure only that screen’s function gets called when needed in the case of multiple screens sharing the same functions.

The buttons are handled mainly by two classes, a NavButton class for constructing a single button, and a Navigation class which creates an arraylist of buttons. When used together it is possible to add as many buttons as we need, and in our case it was three.

In the search screen and the input class, there exists mechanisms for checking whether the user is in the search box or not and adds whatever is typed into the search bar. Clicking the search box would allow the user to enter it, pressing the enter(or return) key or clicking outside the search box quits it, pressing the backspace key removes a character and pressing any other key adds that character to the search box.

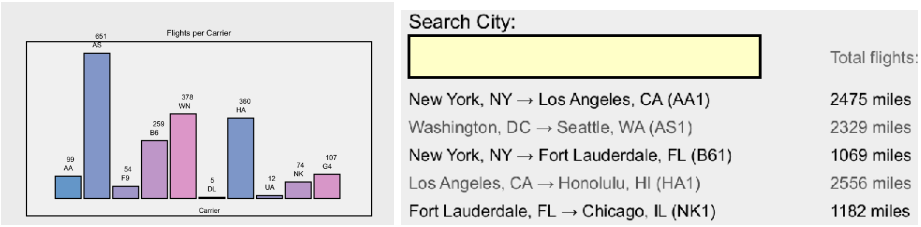
Whenever a character is typed into the search box, a new arraylist would be created and the system would compare the lower case of what is in the search box with the lower case of the origin or the destination city in the dataset. If there is a match, the matching entry will be added to the arraylist and the new arraylist is displayed in place of the previous one.

For the barchartscreen, Used hashmaps to store airport data, sorted it using ArrayLists, and displayed it as a bar chart. Used a mode system to analyse the air traffic both ways.Added interactive controls like a slider, toggle button, and mode buttons using mouse input to allow filtering and updating of the chart.

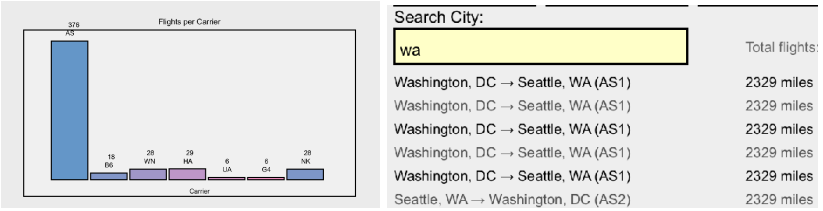
Execution and results:

#Search Screen-

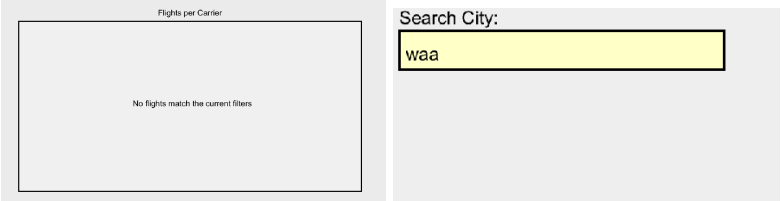
In the search screen, the initial graph looks like this with the full 2k CSV file.



After applying any text input in the search box, the amount of data available to the graph also changes according to the search results. Below is an example of a graph under one of the text inputs.

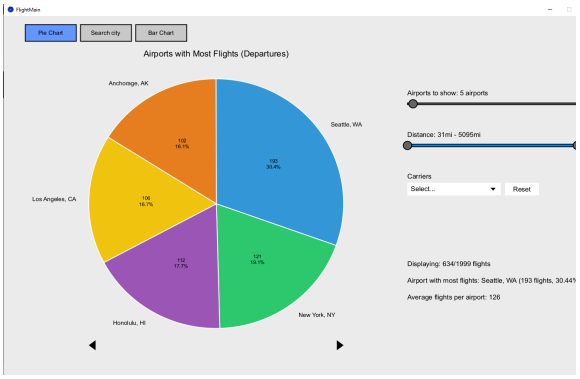


If the user enters something with no matching data, a corresponding message will be displayed, as shown below.

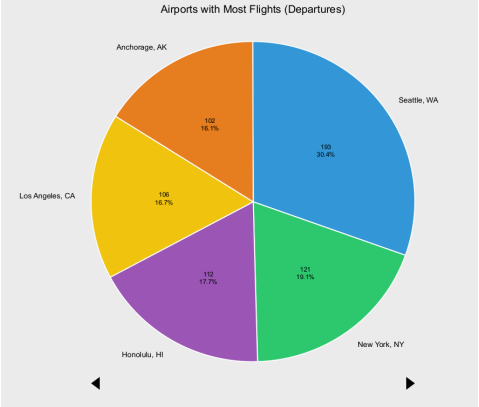


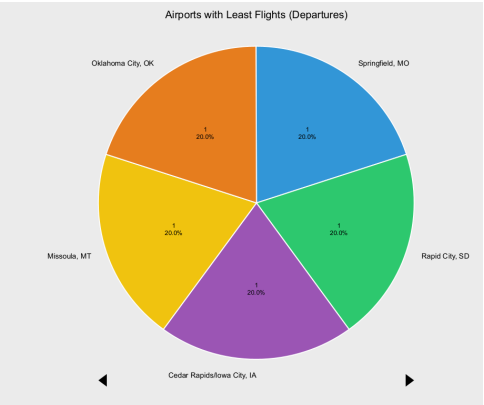
#Pie charts screen-

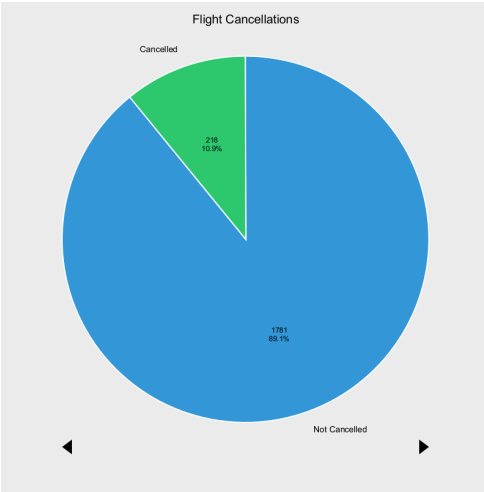
This is what the pie chart screen looks like:

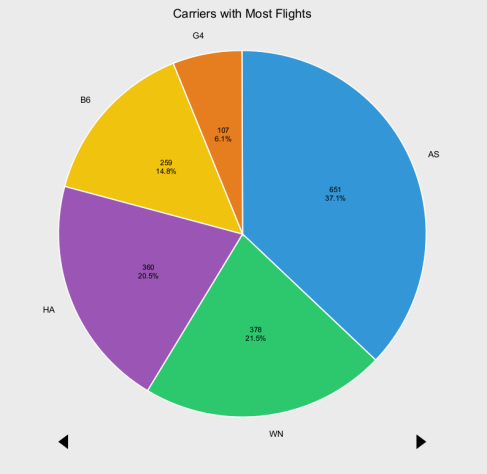


It has a carousel, which contains 4 pie charts displaying different data:





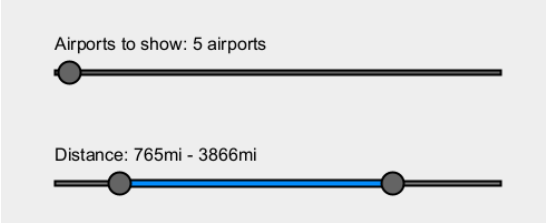




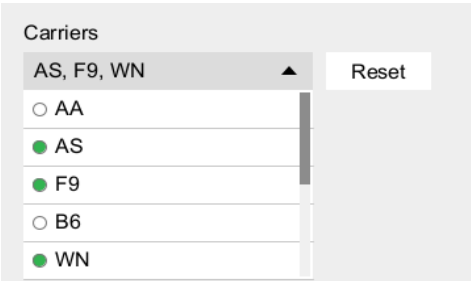
Each pie chart includes data filtering user inputs.

<u>Airports to show</u>: the number of different airports to display on the pie chart (number of slices on the pie chart)

<u>Distance</u>: distance of the flight



<u>Carriers</u>: multi-select input to select the flights of a specific carrier. The reset button clears all selections.



Below the user inputs, there's a section for additional insights from the filtered data. It changes in real-time depending on the user inputs above.

Displaying: 539/1999 flights

Airport with most flights: Seattle, WA (169 flights, 31.35%)

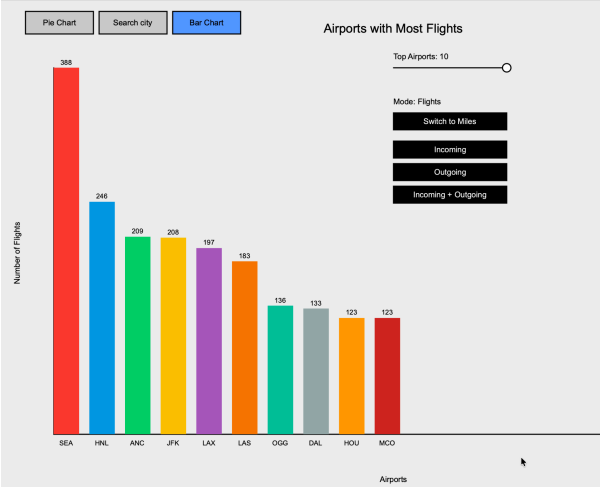
Average flights per airport: 107

Airport with most flight cancellations: Seattle, WA (16)

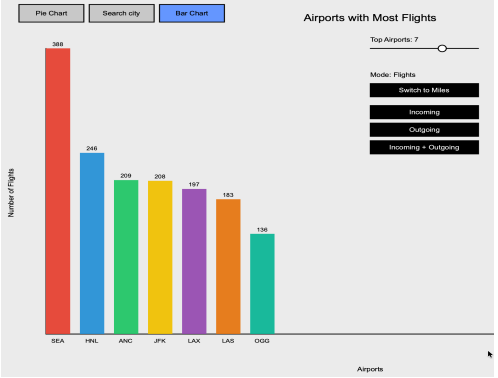
Airport with least flight cancellations: Minot, ND (1)

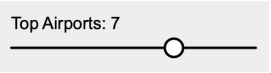
#*Bar charts scree*n-

For the bar charts screen, the initial graph for the number of flights with the 2k CSV file looks like this. It initially displays the top 10 airports with the most flights

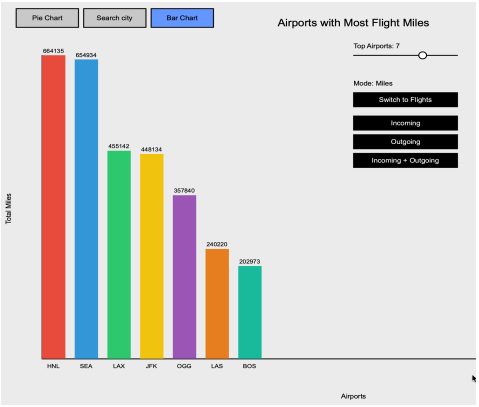


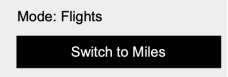
A slider is used to change the amount of bars (airports) are shown on the screen



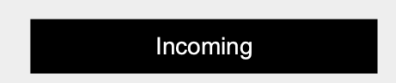


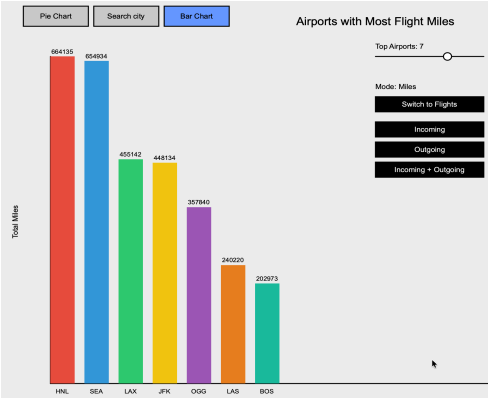
A button is used to change from displaying the number of total fights to the total distance (in flight miles) travelled.

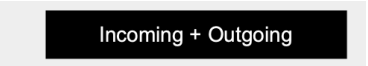




Three more buttons for displaying and filtering between incoming flights only, outgoing flights only and both incoming+outcoming flights.







Conclusion:

This project successfully developed an interactive flight data visualization system using Processing. Key features include CSV file reading, scrollable text, real time search(filtering of data), multiple graphs with sliders and dropdown boxes and screen switching via buttons. Throughout this project, we separated each member's tasks into the three screens, and merged them together towards the end. This approach naturally came to us when we splitted our work, which created quite some trouble when we were in the merging phase.

Future improvements could revolve around working with larger CSV files, such as a loading screen to mask the effects of slowness when loading huge amounts of entries, we had to remove a dataset changing feature because the program gets stuck for quite a long time when trying to work with the 100k CSV file and above, and we apologise for that. Our UI currently looks quite plain too with a completely white background and let's not forget about better code management with GitHub to avoid painful code merges.

Contributions:

Xianren Liu:

- Flight class, NavButton class, Navigation Class, SearchScreen class, Text Input extension of the Input class, minor changes to the PieChartScreen class and BarChartsScreen, changes to FlightsMain in relation to the classes above

Jayden Qin:

- Created classes: Flights, Input, Slider, RangeSlider, Dropdown, Button, MultiSelectDropdown, Widget, Carousel, ChartData, Chart, PieChart, PieChartsScreen
- Created screen: Pie Charts Screen
- Organised and managed GitHub repository/merging team code (we used to work in individual workspaces)
- Responsible for refactoring the code

Abdul Rashid:

- Created screen: barChartsScreen
- implement the arrow scroll on SearchScreen class
- helped with the merging of the teams code
- drafted rough plan and helped assign tasks each week
- edited vid

Git history:

Commits on Apr 10, 2026

Update jayden-qin committed 9 minutes ago		57d6f7d
Added comments explaining code jayden-qin committed 15 minutes ago		889f62f
Update SearchScreen.pde Liux18 authored 37 minutes ago	Verified	fcc6b13
Update PieChartsScreen.pde Liux18 authored 38 minutes ago	Verified	0c24eaa
Update barChartsScreen.pde Liux18 authored 38 minutes ago	Verified	ddbd294
Update FlightMain.pde Liux18 authored 39 minutes ago	Verified	c71dac7
Update barChartsScreen.pde Liux18 authored 45 minutes ago	Verified	2a44fad
Update PieChartsScreen.pde Liux18 authored 46 minutes ago	Verified	4cf157f
Update SearchScreen.pde Liux18 authored 47 minutes ago	Verified	cf1e914
Update Input.pde Liux18 authored 49 minutes ago	Verified	4681c94
Update Button.pde Liux18 authored 50 minutes ago	Verified	3e85ed1
Update Button.pde Liux18 authored 1 hour ago	Verified	22dc2d2
Update SearchScreen.pde Liux18 authored 1 hour ago	Verified	852b001
Removed redundant comments jayden-qin committed 1 hour ago		85770ac

Update barChartsScreen.pde Liux18 authored 5 hours ago	Verified	f4f0534
Update PieChartsScreen.pde Liux18 authored 5 hours ago	Verified	433874f
Update Button.pde Liux18 authored 5 hours ago	Verified	d28bc41
Update Input.pde Liux18 authored 5 hours ago	Verified	bda55ff
Update SearchScreen.pde Liux18 authored 5 hours ago	Verified	fd901b1

Commits on Apr 9, 2026

Revise project tasks in README Jayden-Q authored yesterday	Verified	30d0f56
Update README rashidabtcd authored yesterday	Verified	dd61a9b
Update barChartsScreen.pde rashidabtcd authored yesterday	Verified	c96f0cd
Update barChartsScreen.pde rashidabtcd authored yesterday	Verified	1286ae4
added comments rashidabtcd authored yesterday	Verified	14a31be
Update barChartsScreen.pde rashidabtcd authored yesterday	Verified	8e6a10d
added incoming and out coming rashidabtcd authored yesterday	Verified	6f604ba
Added new inputs + added metrics jayden-qin committed 2 days ago		6f084ef

Commits on Apr 8, 2026

Fixed conflicts jayden-qin committed 2 days ago	35e189e
Merged development into fixes branch and resolved conflicts jayden-qin committed 2 days ago	a384ab6
Merge development into fixes branch + fix conflicts jayden-qin committed 2 days ago	b087581
Save local work before syncing jayden-qin committed 2 days ago	334b076

Commits on Apr 2, 2026

Update FlightMain.pde Liux18 authored last week	Verified	b684b5a
Add comment for bar chart screen class rashidabtcd authored last week	Verified	f74d568
Refactor BarChartsScreen for clarity and functionality rashidabtcd authored last week	Verified	5fa30f1
done bar char rashidabtcd authored last week	Verified	8f0c5b4

Commits on Apr 2, 2026

Update Button.pde Liux18 authored last week	Verified	a43428a
Update SearchScreen.pde Liux18 authored last week	Verified	e502fce
Update SearchScreen.pde Liux18 authored last week	Verified	c5b5e46
Update FlightMain.pde Liux18 authored last week	Verified	b1e306d

Commits on Apr 1, 2026

Delete flights/HomeScreen.pde Liux18 authored last week	Verified	f00ffcc
Delete Button.pde Liux18 authored last week	Verified	71d9846
Create Button.pde Liux18 authored last week	Verified	36dd737
Add files via upload Liux18 authored last week	Verified	878bb7d
Update Widget.pde Liux18 authored last week	Verified	aeb4454
Update Input.pde Liux18 authored last week	Verified	401c32d
Update FlightMain.pde Liux18 authored last week	Verified	315e125
Update PieChartsScreen.pde Liux18 authored last week	Verified	fb8c14e

Update SearchScreen.pde Liux18 authored last week	Verified	99179cf
Update barChartsScreen.pde Liux18 authored last week	Verified	7488816
Update SearchScreen.pde Liux18 authored last week	Verified	1cb492a
Update barChartsScreen.pde Liux18 authored last week	Verified	61569fa

Commits on Mar 31, 2026

Added home screen + buttons to navigate screens + fixed input interactions bugs jayden-qin committed last week	3a3ed57

Commits on Mar 28, 2026

Merged code (bar + pie charts screens) jayden-qin committed 2 weeks ago	f0f8e10

Commits on Mar 23, 2026

Added class Input jayden-qin committed 3 weeks ago	e1f3cfd

Commits on Mar 19, 2026

updated contributions Liux18 authored 3 weeks ago	Verified	61b487a
Update README Liux18 authored 3 weeks ago	Verified	454d346
Update FlightMain.pde Liux18 authored 3 weeks ago	Verified	92ab2db

Add comment and initialize dateInput and yearInput rashidabtcd authored 3 weeks ago	Verified	8572cbd
Update FlightMain.pde Liux18 authored 3 weeks ago	Verified	e4d6005
Update FlightMain.pde Liux18 authored 3 weeks ago	Verified	058d9a7
Update FlightMain.pde Liux18 authored 3 weeks ago	Verified	8cd4a5e
Add files via upload rashidabtcd authored 3 weeks ago	Verified	482f4f9
Update FlightMain.pde Liux18 authored 3 weeks ago	Verified	2a3223e
Update FlightMain.pde Liux18 authored 3 weeks ago	Verified	0f4a6be
Rename Flight.pde to Flight_Var.pde Liux18 authored 3 weeks ago	Verified	d1677cc
Refactor jayden-qin committed 3 weeks ago		6747f64
Refactor jayden-qin committed 3 weeks ago		6cd036f
Revise week 9 contributions in README rashidabtcd authored 3 weeks ago	Verified	2f38d74
draft plan for project- Abdul rashidabtcd authored 3 weeks ago	Verified	795e320
Update flight_WIP Liux18 authored 3 weeks ago	Verified	f8436f8

Commits on Mar 19, 2026

Update flight_WIP Liux18 authored 3 weeks ago	Verified	3720dd5
flight_WIP Liux18 authored 3 weeks ago	Verified	fc50902

Commits on Mar 18, 2026

Merge pull request #2 from Jayden-Q/development Jayden-Q authored 3 weeks ago	Verified	887c0d3
Refactor jayden-qin committed 3 weeks ago		c2d8e39

Commits on Mar 12, 2026

Update project tasks for week 8 rashidabtcd authored last month	Verified	1cdf64d
Update README with project planning details rashidabtcd authored last month	Verified	68dce46
Add files via upload Liux18 authored last month	Verified	38b125a
Update project roles in README rashidabtcd authored last month	Verified	1970f9b
Update README rashidabtcd authored last month	Verified	55cd1d3
Add files via upload Liux18 authored last month	Verified	49aac53
Update README to include week9 rashidabtcd authored last month	Verified	1060091
Update README with week 8 contributions rashidabtcd authored last month	Verified	7bc26eb

Merge pull request #1 from Jayden-Q/development Jayden-Q authored last month	Verified	f6ab572
Implemented reading data from csv file jayden-qin committed last month		067f986
Add files via upload Liux18 authored last month	Verified	1f0a36c
Added team members to README jayden-qin committed last month		cc43327
Create README Jayden-Q authored last month	Verified	b2852f9